

Introduction to Linear Algebra

ECON 5753 — University of Arkansas

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Matrix Fundamentals

Matrix Transformations

Inverse of a Matrix

Normal Random Variables

Derivatives of matrix expressions

Data

The basic currency of *forecasting* is **data**. Data typically looks like:

Columns are variables ↓

	mpg	cyl	hp	wt	am	gear
<i>Rows are observations →</i>	21.0	6	110	2.620	1	4
	21.0	6	110	2.875	1	4
	22.8	4	93	2.320	1	4
	21.4	6	110	3.215	0	3
	18.7	8	175	3.440	0	3

Data as a matrix

The basic currency of *linear algebra* are **matrices**. Matrices look like this:

$$\begin{bmatrix} 21.0 & 6 & 110 & 2.620 & 1 & 4 \\ 21.0 & 6 & 110 & 2.875 & 1 & 4 \\ 22.8 & 4 & 93 & 2.320 & 1 & 4 \\ 21.4 & 6 & 110 & 3.215 & 0 & 3 \\ 18.7 & 8 & 175 & 3.440 & 0 & 3 \end{bmatrix}$$

Matrix notations

$$\begin{bmatrix} 21.0 & 6 & 110 & 2.620 & 1 & 4 \\ 21.0 & 6 & 110 & 2.875 & 1 & 4 \\ 22.8 & 4 & 93 & 2.320 & 1 & 4 \\ 21.4 & 6 & 110 & 3.215 & 0 & 3 \\ 18.7 & 8 & 175 & 3.440 & 0 & 3 \end{bmatrix}$$

Let's call this matrix, $\mathbf{A}$. Each element of $\mathbf{A}$ can be referred to as $A_{i,j}$ where i is the *row* and j is the *column*.

Matrix notations

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→ Unit i 's data is written $A_{i, \cdot}$.

→ Variable j is written $A_{\cdot, j}$

Matrix Fundamentals

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

The **Size/Dimension** a matrix: (rows) $\times$ (columns). E.g. $\mathbf{A}$ is a 2×3 matrix.

→ The element in row i and column j is referred to as a_{ij} or A_{ij} .

→ A **square matrix** is one with the same number of rows and columns.

Vectors

A **column vector**, often just “vector”, is a $N \times 1$ matrix:

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

In our previous discussion, each variable in a dataset can be thought of as a *column vector*. A *matrix*, then, is a collection of variables (vectors) glued together

Matrix Addition and Subtraction

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \text{ and } B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix}$$

Dimensions must match:

$$(r \times c) \pm (r \times c) \implies (r \times c)$$

A and B are both 2×3 matrices, so we can add and subtract them:

$$A \pm B = \begin{bmatrix} a_{11} \pm b_{11} & a_{12} \pm b_{12} & a_{13} \pm b_{13} \\ a_{21} \pm b_{21} & a_{22} \pm b_{22} & a_{23} \pm b_{23} \end{bmatrix}$$

Scalar Multiplication

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

For any scalar c (a number like 2 or -4):

$$cA = \begin{bmatrix} c * a_{11} & c * a_{12} & c * a_{13} \\ c * a_{21} & c * a_{22} & c * a_{23} \end{bmatrix}$$

Scalar Multiplication

This is particularly useful when working with vectors,

$$\beta v = \begin{bmatrix} \beta v_1 \\ \beta v_2 \\ \vdots \\ \beta v_N \end{bmatrix}$$

Using matrices

the most popular forecasting method we will learn is the *linear regression*. Say we want to predict mpg given other variables about the car (hp and wt).

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the most popular forecasting method we will learn is the *linear regression*. Say we want to predict mpg given other variables about the car (hp and wt).

To do so, we take variables, $A_{.,j}$, and multiply it by their slope parameter to make predictions:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \alpha + \begin{bmatrix} 110 \\ 110 \\ 93 \\ 110 \\ 175 \end{bmatrix} \beta_{\text{hp}} + \begin{bmatrix} 2.620 \\ 2.875 \\ 2.320 \\ 3.215 \\ 3.440 \end{bmatrix} \beta_{\text{wt}}$$

Matrix multiplication

Using our rules of scalar multiplication and addition of vectors, we get:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \alpha + \begin{bmatrix} 110 \\ 110 \\ 93 \\ 110 \\ 175 \end{bmatrix} \beta_{\text{hp}} + \begin{bmatrix} 2.620 \\ 2.875 \\ 2.320 \\ 3.215 \\ 3.440 \end{bmatrix} \beta_{\text{wt}} = \begin{bmatrix} \alpha + 110\beta_{\text{hp}} + 2.620\beta_{\text{wt}} \\ \alpha + 110\beta_{\text{hp}} + 2.875\beta_{\text{wt}} \\ \alpha + 93\beta_{\text{hp}} + 2.320\beta_{\text{wt}} \\ \alpha + 110\beta_{\text{hp}} + 3.215\beta_{\text{wt}} \\ \alpha + 175\beta_{\text{hp}} + 3.440\beta_{\text{wt}} \end{bmatrix}$$

This is a *linear combination* of the columns of the matrix

Matrix Multiplication

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \text{ and } D = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \\ d_{31} & d_{32} \end{bmatrix}$$

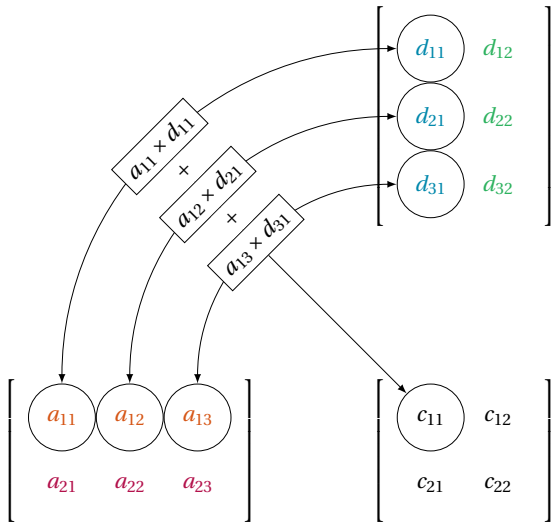
Inner Dimensions must match:

$$(r \times \underline{c}) \times (\underline{c} \times p) \implies (r \times p)$$

A is a 2×3 and D is a 3×2 matrix, so we can multiply (the 2s are equal):

$$A \times D = \begin{bmatrix} a_{11}d_{11} + a_{12}d_{21} + a_{13}d_{31} & a_{11}d_{12} + a_{12}d_{22} + a_{13}d_{32} \\ a_{21}d_{11} + a_{22}d_{21} + a_{23}d_{31} & a_{21}d_{12} + a_{22}d_{22} + a_{23}d_{32} \end{bmatrix}$$

D : 3 rows 2 columns



A : 2 rows 3 columns

C = A $\times$ B : 2 rows 3 columns

Matrix Multiplication

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \text{ and } D = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \\ d_{31} & d_{32} \end{bmatrix}$$

Each element of the resulting $A \times D$ is the vector **dot product** between each *row* of A and each column of D :

$$(A \times D)_{i,j} = A_{i,\cdot} \cdot D_{\cdot,j}$$

Matrix Multiplication Practice

$$A = \begin{bmatrix} 1 & 4 & 1 \\ 2 & 3 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 2 & 2 \end{bmatrix}$$

What is $A \times B$? What is $B \times A$?

Matrix Multiplication in Regression

It turns out, we can write out our linear combination as a matrix times a vector

$$\begin{bmatrix} 1 & 110 & 2.620 \\ 1 & 110 & 2.875 \\ 1 & 93 & 2.320 \\ 1 & 110 & 3.215 \\ 1 & 175 & 3.440 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta_{\text{hp}} \\ \beta_{\text{wt}} \end{bmatrix}$$

Verify that this product creates the same result

Why call it “multiplication”

There are three rules we learn about regular multiplication:

1. associative: $(ab)c = a(bc)$;
2. distributive: $a(b + c) = ab + ac$; and
3. commutative: $ab = ba$

Why call it “multiplication”

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1. associative: $(ab)c = a(bc)$;
2. distributive: $a(b + c) = ab + ac$; and
3. commutative: $ab = ba$

Matrices with multiplication and addition are associative, distributive, but *NOT* commutative:

1. $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
2. $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$
3. However, $\mathbf{AB} \neq \mathbf{BA}$ (in some cases this holds, but not generally)

Commutative

$$\mathbf{AB} \neq \mathbf{BA}$$

In some cases, one of these two might not even exist

For example, E.g. $\mathbf{A}$ is a 3×2 matrix and $\mathbf{B}$ is a 2×1 matrix.

→ $\mathbf{AB}$ is well-defined (since the number of columns of $\mathbf{A}$ = the number of rows of $\mathbf{B}$)

→ But, $\mathbf{BA}$ is not well-defined

Identity Matrix

In the same way that the number 1 is special in multiplication of numbers (the ‘identity’), the **identity matrix** takes the following form:

$$\mathbb{I}_n = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

The identity matrix has the property that $\mathbb{I}_n \mathbf{A} = \mathbf{A} \mathbb{I}_n = \mathbf{A}$

Uses of Matrix Multiplication

Say I have some variable in my dataset x and I want to know the sum of x (or $1/n$ * the sum to get the sample mean).

Take a few moments and try and think about what the matrix S would need to be to calculate the sample mean of x :

$$\frac{1}{n} \mathbf{S} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \frac{1}{n} (x_1 + \cdots + x_n)$$

Uses of Matrix Multiplication

$$\frac{1}{n} \begin{bmatrix} 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \frac{1}{n} (x_1 + \dots + x_n)$$

Answer: The **row-vector** consisting of all 1s

Uses of Matrix Multiplication

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Answer: The **row-vector** consisting of all 1s

The column vector consisting of all 1s is often called “iota”, $\iota = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$.

The row vector is just ι flipped on it's side

Transpose

The **transpose** of a vector will do this “flipping” of vectors and matrices. We denote the transpose as $v^\top$ or v'

→ The latter can be confused with ‘derivative’, but is easier to write and very common.

For vectors, the transpose is

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \iff x' = \begin{bmatrix} x_1 & \dots & x_n \end{bmatrix}$$

Transpose

For matrices, the transpose is

$$\mathbf{A} = \begin{bmatrix} A_{.,1} & \vdots & A_{.,K} \end{bmatrix} \iff \mathbf{A}' = \begin{bmatrix} A'_{.,1} \\ \dots \\ A'_{.,K} \end{bmatrix}$$

The i -th column of $\mathbf{A}$ becomes the i -th row of $\mathbf{A}'$

Transpose rules

One rule worth knowing is $(ab)' = b'a'$

→ I kind of remember it since transposing “flips” the rows and columns, you also flip the order of the vectors/matrices.

Sample mean

With this, we can write out sample mean more simply as

$$\bar{x} = \frac{1}{n} \iota' x$$

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Or, if we wanted the means of multiple variables, we could put them in a matrix and

$$\frac{1}{n} \iota' \mathbf{X}$$

would be a row-vector of sample means

Sample variance

Recall from your introductory statistics course, the sample variance of a variable is given by

$$s^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2$$

Let $\tilde{x}_i \equiv x_i - \bar{x}$. For a minute, think about how you might calculate the sum of squares using the vector $\tilde{x}$

Sample variance

$$s^2 = \frac{1}{n-1} \tilde{x}' \tilde{x} = \frac{1}{n-1} \begin{bmatrix} \tilde{x}_1 & \dots & \tilde{x}_n \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_n \end{bmatrix}$$

Sample covariance

Similarly, we can write the covariance of two-variables as

$$\frac{1}{n-1} \tilde{x}' \tilde{y} = \frac{1}{n-1} \sum_i \tilde{x}_i \tilde{y}_i$$

Variance Covariance Matrix

Consider a matrix of variable where each column is a (de-meaned) sample.

$$\tilde{\mathbf{X}} = \begin{bmatrix} x_1 - \bar{x} & y_1 - \bar{y} & z_1 - \bar{z} \\ x_2 - \bar{x} & y_2 - \bar{y} & z_2 - \bar{z} \\ \vdots & \vdots & \vdots \\ x_n - \bar{x} & y_n - \bar{y} & z_n - \bar{z} \end{bmatrix},$$

where $\bar{x}$ is the mean of variable x .

Variance Covariance Matrix

The Variance-Covariance Matrix is $\frac{1}{n-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{X}}$

$$= \frac{1}{n-1} \begin{bmatrix} \sum_{i=1}^n (x_i - \bar{x})^2 & \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) & \sum_{i=1}^n (x_i - \bar{x})(z_i - \bar{z}) \\ \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) & \sum_{i=1}^n (y_i - \bar{y})^2 & \sum_{i=1}^n (y_i - \bar{y})(z_i - \bar{z}) \\ \sum_{i=1}^n (z_i - \bar{z})(x_i - \bar{x}) & \sum_{i=1}^n (z_i - \bar{z})(y_i - \bar{y}) & \sum_{i=1}^n (z_i - \bar{z})^2 \end{bmatrix}$$

$$= \begin{bmatrix} \text{Var}(x) & \text{Cov}(x, y) & \text{Cov}(x, z) \\ \text{Cov}(y, x) & \text{Var}(y) & \text{Cov}(y, z) \\ \text{Cov}(z, x) & \text{Cov}(z, y) & \text{Var}(z) \end{bmatrix}$$

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Matrix Times a Vector (Transformations)

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \text{ and } x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

An $n \times n$ matrix, A , times a $n \times 1$ vector, x , is a transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$. So A takes x , rotates it around and/or shrinks or extends the line.

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In general,

$$Ax = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix} \in \mathbb{R}^2$$

Transformation Examples

Our identity matrix, is a boring transformation:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 * 2 + 0 * 3 \\ 0 * 2 + 1 * 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Transformation Examples

Reflection on the Y-axis:

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$$

Transformation Examples

Reflection 90 degrees clockwise:

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ -x \end{bmatrix}$$

Transformation Examples

Enlargement by scale factor a in the x direction and scale factor b in the y direction:

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax \\ by \end{bmatrix}$$

Combination of Transformations

Let's say I want to rotate a vector 90 degrees clockwise and then keep only the x direction (i.e. scale the y by 0.)

→ I just multiply the matrices in the order I want to do them:

Let's try:

$$\underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{\text{Keep only } x} \overbrace{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}}^{\text{Rotate 90 degrees clockwise}} \begin{bmatrix} x \\ y \end{bmatrix} =$$

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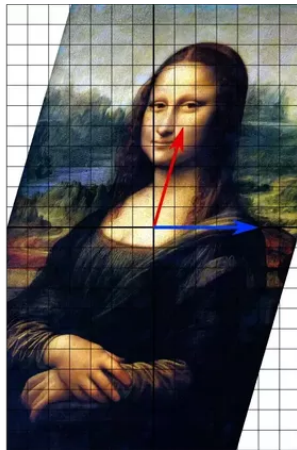
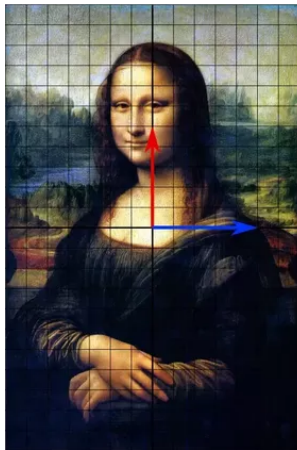
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Determinant of a Matrix

When we rotate and scale an image, we are just doing many many vectors times a transformation matrix.

The **determinant** asks how much does the area change with our transformation:



Formula for 2×2 Determinant

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

The **Determinant** of A is given by:

$$\det(A) = a_{11} * a_{22} - a_{12} * a_{21}$$

- If $\det(A) = 1$, then the transformation preserves area
- If $\det(A)$ is greater than/smaller than 1, then the transformation grows/shrinks area.
- If $\det(A) = 0$, then the transformation area shrinks to zero (you lose dimensions).

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Matrix Inverse

The final necessary linear algebra concept is that of the **matrix inverse**.

→ It is equivalent to $1/x * x = 1$

A *square matrix* (i.e., dimension $n \times n$) $\mathbf{A}$ has an inverse $\mathbf{A}^{-1}$ if

$$\mathbf{A}^{-1}\mathbf{A} = \mathbb{I}_n = \mathbf{A}\mathbf{A}^{-1}$$

Inverse of Matrix

$$\mathbf{A}^{-1}\mathbf{A}x = \mathbb{I}_n x = x$$

The inverse of a matrix "undoes" the transformation done by A , i.e

$$AA^{-1}x = A^{-1}Ax = x.$$

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$$AA^{-1}x = A^{-1}Ax = x.$$

If the determinant of a matrix is 0, then the transformation does not have an inverse.

→ For example, the matrix that only keeps the x component can't be inverted (what is the correct y value?)

Inverse of 2x2 Matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

For 2x2 matrices, there is a nice formula for the inverse:

$$A^{-1} = \frac{1}{\underbrace{a_{11}a_{22} - a_{12}a_{21}}_{=\det(A)}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

Inverse Example

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ -4 & 10 \end{bmatrix}$$

Find the inverse of $\mathbf{A}$ and verify it is indeed the inverse of $\mathbf{A}$

Inverse Example

$$\mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

What is the determinant of $\mathbf{B}$? Does $\mathbf{B}$ have an inverse?

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Repeated Sampling

When conducting forecasts, we want to express uncertainty around our best guess, what is called **statistical inference**.

We use the **repeated sampling** framework to think about this:

- We collect one sample of data and estimate a model
- Imagine collecting *many many* samples in the same way and estimating the model for each sample.

Repeated Sampling and the Normal Distribution

In almost all cases, our estimates will be **normally distributed** (at least in large samples).

For example, we know that the sample mean of a variable is approximately normally distributed:

$$\bar{x} = \frac{1}{n} \sum_i x_i \sim \mathcal{N}\left(\mu, \frac{\sigma_x^2}{n}\right)$$

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This allowed us to do things like:

1. Form confidence intervals, and
2. Perform hypothesis tests

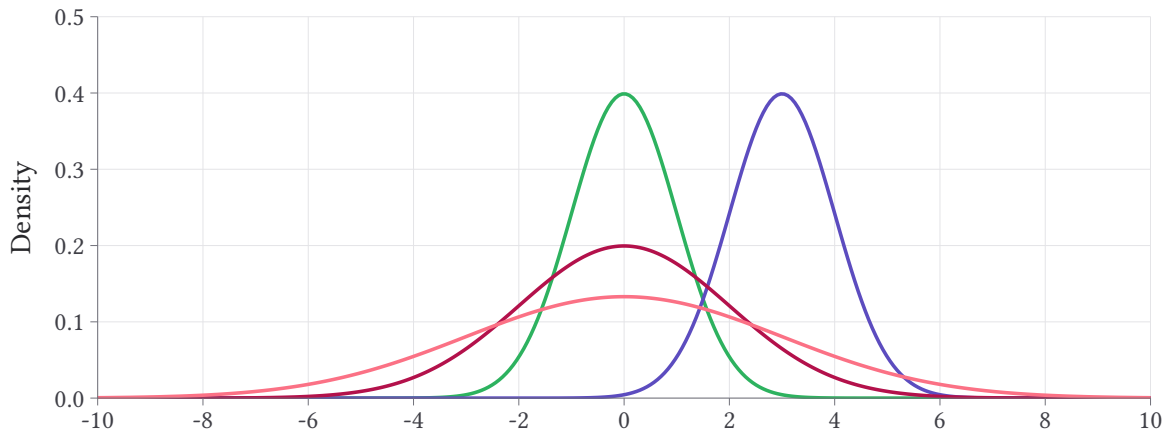
Normal Distribution

We say a random variable is normally distributed and write it as

$$x \sim \mathcal{N}(\mu, \sigma^2),$$

where $\mu = \mathbb{E}[x]$ and $\sigma^2 = \text{Var}(x)$

PDFs: $Z = \mathcal{N}(0, 1)$, $\mathcal{N}(3, 1)$, $\mathcal{N}(0, 4)$, $\mathcal{N}(0, 9)$



Properties of Normal Distribution

We have $x \sim \mathcal{N}(\mu, \sigma^2)$. Say we perform some transformation of x :

$$ax + b$$

The modified variable is still normally distributed. Using properties of expectation and variance, we have:

1. $\mathbb{E}[ax + b] = a\mathbb{E}[x] + b = a\mu + b$
2. $\text{Var}(ax + b) = \text{Var}(ax) = a^2 \text{Var}(x) = a^2\sigma^2$

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$$ax + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$$

Random Vectors

We can extend this distribution to the **joint distribution** of multiple random variables. Let

$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ be a **random vector**.

These variables could be correlated with one another. For example, say x_1 is height and x_2 is weight of a surveyed person (or a sample mean of them).

→ In repeated sampling, we expect x_1 and x_2 to be positively correlated.

Expectation of Random Vector

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

The expectation of the vector is just the expectation of each component variable

$$\mathbb{E}[x] = \begin{bmatrix} \mathbb{E}[x_1] \\ \vdots \\ \mathbb{E}[x_n] \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix}$$

Covariance of Random Vector

$$\tilde{x} = \begin{bmatrix} x_1 - \mu_1 \\ \vdots \\ x_n - \mu_n \end{bmatrix}$$

This vector has mean 0. We can write the covariance of the random vector as

$$\mathbb{E}[\tilde{x}\tilde{x}'] = \begin{bmatrix} \mathbb{E}[(x_1 - \mu_1)(x_1 - \mu_1)] & \dots & \mathbb{E}[(x_1 - \mu_1)(x_n - \mu_n)] \\ \vdots & \ddots & \vdots \\ \mathbb{E}[(x_n - \mu_n)(x_1 - \mu_1)] & \dots & \mathbb{E}[(x_n - \mu_n)(x_n - \mu_n)] \end{bmatrix}$$

Covariance of Random Vector

From the previous slide, we have

$$\Sigma \equiv \mathbb{E}[(x - \mu)(x - \mu)'] = \begin{bmatrix} \text{Var}(x_1) & \dots & \text{Cov}(x_1, x_n) \\ \vdots & \ddots & \vdots \\ \text{Cov}(x_n, x_1) & \dots & \text{Var}(x_n) \end{bmatrix}$$

The diagonal of this matrix is the variance of each variable. The off-diagonal elements are the covariance between the variables.

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The diagonal of this matrix is the variance of each variable. The off-diagonal elements are the covariance between the variables.

→ If the off-diagonal elements are all 0, then the random variables are uncorrelated.

Multivariate Normal Distribution

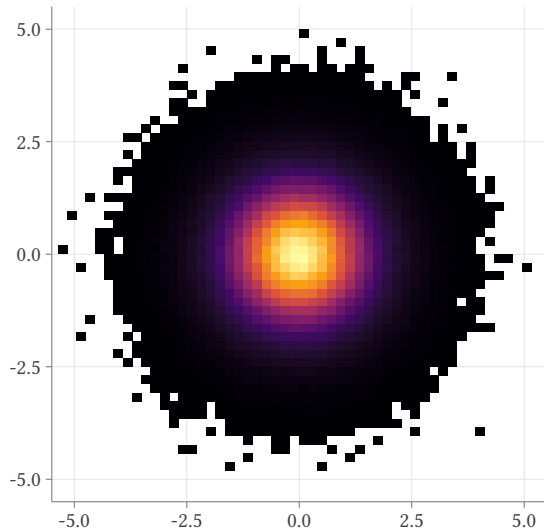
In particular, we might estimate a few parameters in our model and they will each be normally distributed.

But, even stronger, these estimates (a random vector) x are multivariate normally distributed:

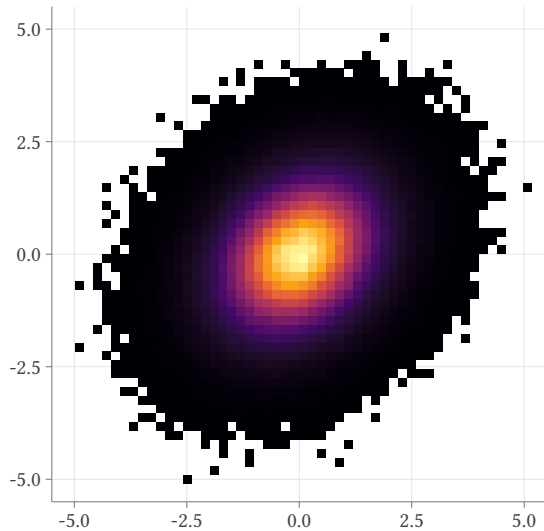
$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \sim \mathcal{N}(\mu, \Sigma)$$

where $\Sigma \equiv \mathbb{E}[(x - \mu)(x - \mu)']$

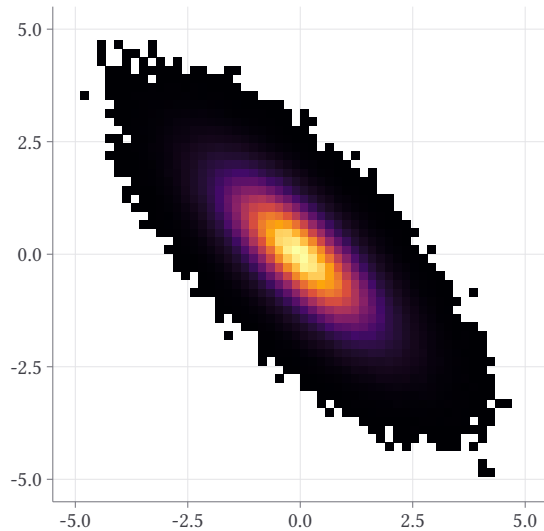
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$$



$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.25 \\ 0.25 & 1 \end{bmatrix}\right)$$



$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & -0.75 \\ -0.75 & 1 \end{bmatrix}\right)$$



Linear combinations of multivariate random variable

Say $x \sim \mathcal{N}(\mu, \Sigma)$ and $a = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$

Then, the product $a'x$ is a linear combination of x_i and is normally distributed:

$$a'x = \sum_i a_i x_i \sim \mathcal{N}(?, ?)$$

Linear combinations of multivariate random variable

Say $x \sim \mathcal{N}(\mu, \Sigma)$ and $a = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$

First, consider the expectation of $a'x$

$$\begin{aligned}\mathbb{E}[a'x] &= \mathbb{E}\left[\sum_i a_i x_i\right] = \sum_i a_i \mathbb{E}[x_i] \\ &= \sum_i a_i \mu_i \\ &= a' \mu\end{aligned}$$

Linear combinations of multivariate random variable

Second, consider the variance of $a'x$

$$\begin{aligned}\text{Var}(a'x) &= \mathbb{E}[(a'x - a'\mu)(a'x - a'\mu)'] \\ &= \mathbb{E}[a'(x - \mu)(x - \mu)']\end{aligned}$$

Linear combinations of multivariate random variable

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$$\begin{aligned}\text{Var}(a'x) &= \mathbb{E}[(a'x - a'\mu)(a'x - a'\mu)'] \\ &= \mathbb{E}[(a'(x - \mu))(a'(x - \mu))'] \\ &= \mathbb{E}[a'(x - \mu)(x - \mu)'a] = a'\Sigma a\end{aligned}$$

Linear combinations of multivariate random variable

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Therefore, if $x \sim \mathcal{N}(\mu, \Sigma)$, then

$$a'x = \sum_i a_i x_i \sim \mathcal{N}(a'\mu, a'\Sigma a)$$

Matrix Fundamentals

Matrix Transformations

Inverse of a Matrix

Normal Random Variables

Derivatives of matrix expressions

Derivatives

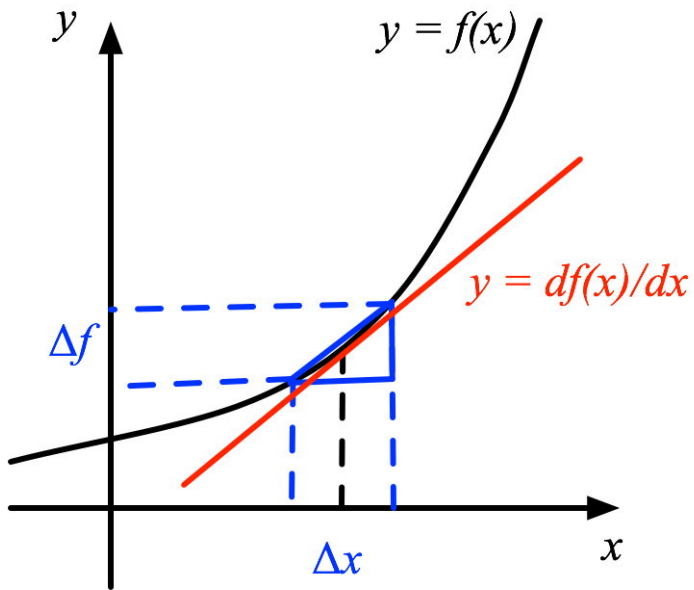
If you recall, we had the following definition of a derivative:

$$\frac{d}{dx}f(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

You could write this as an approximation:

$$f(x+dx) - f(x) \approx \frac{d}{dx}f(x) * dx$$

In words, the change in f is approximately equal to the derivative times the change in x



Taylor Expansion

This formulation is actually the basis of the Taylor Expansion that you may have learned in your calculus course:

$$f(x + dx) \approx f(x) + \frac{d}{dx}f(x) * dx + \text{something small}$$

→ So long as dx is “small”, this approximation is quite accurate.

Multivariate derivative

Say f is now a function that takes a vector x ($\mathbb{R}^n$) and produces a scalar output ($\mathbb{R}$). How f changes depends on which of the input x_i you are changing.

$$\frac{\partial}{\partial x} f(x) \equiv \begin{bmatrix} \frac{\partial}{\partial x_1} f(x) \\ \vdots \\ \frac{\partial}{\partial x_n} f(x) \end{bmatrix}$$

This is called the “gradient” and it is a vector containing the partial derivatives

Multivariate derivative

Our linear approximation holds in this setting to. Let dx be the column vector of changes in each x_i : $dx = \begin{bmatrix} dx_1 & \dots & dx_n \end{bmatrix}'$.

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Useful matrix derivative rules

Let x and y be vectors of the same length. The derivative of the dot product $x'y$ can be given as

$$\frac{\partial}{\partial x} x'y = y$$

and

$$\frac{\partial}{\partial y} x'y = x$$

Useful matrix derivative rules

Let $\mathbf{A}$ be a matrix and x a vector.

$$\frac{\partial}{\partial x} \mathbf{A}x = \mathbf{A}$$

More, using the *chain rule* for derivatives, you could show

$$\frac{\partial}{\partial x} x' \mathbf{A}x = 2\mathbf{A}x$$

You can see the proofs here: <https://bookdown.org/compfinezbook/introcompfinr/Derivatives-of-Simple-Matrix-Functions.html>